



Stern Gerlach Experiment :

The direct evidence for existence of mag. moments of atoms and their space quantization is provided by the experiment of Stern & Gerlach & provides a direct-experimental proof that an electron has an intrinsic magnetic moment and a spin quantum no of $\frac{1}{2}$.

The experiment measures the magnetic moment of an atom as a whole. The atoms are chosen for the experiment which in the ground state have zero orbital momentum. The entire magnetic moment of the atom could then be attributed to the spin of its electron. Atoms of Hydrogen H , Na, K, Cu, Ag, Au were used.

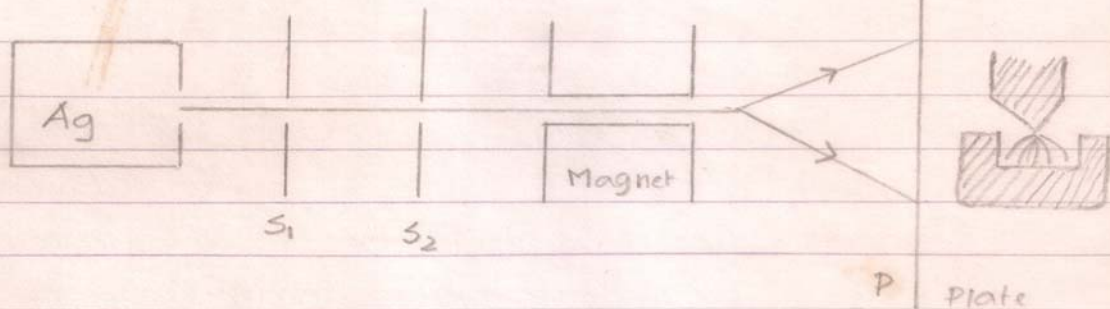
In such atoms the movement of electron is equivalent to a current flowing in a circular loop of wire and has the corresponding magnetic properties. In particular it behaves as a magnetic dipole which experiences equal but opposite forces when placed in a uniform magnetic field and therefore the resultant force is zero and the atom does not experience any displacement.

However when the atom is placed in non-uniform field, the atom experiences a resultant force proportional to the gradient

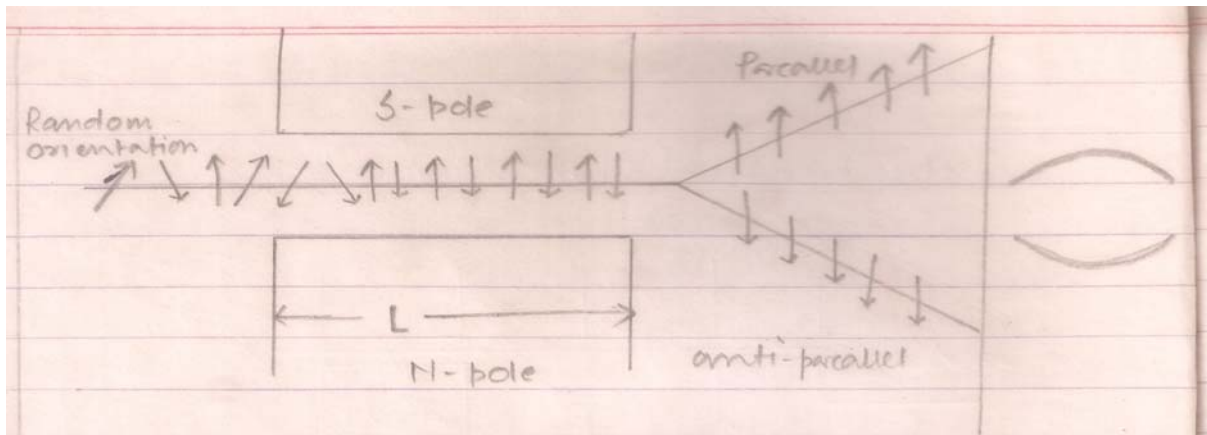


of the field.

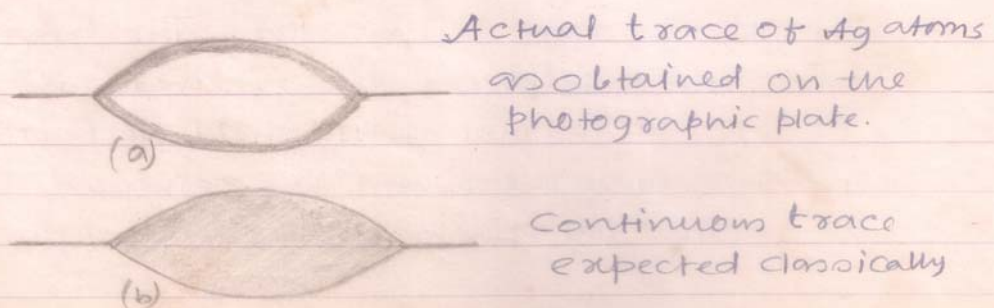
Experimental Arrangement :



In the original experiment silver was heated in an oven and a stream of silver atoms having velocities corresponding to the oven temp. emerged. Two slits S_1 & S_2 limited the stream to a very fine pencil which was then allowed to pass between the poles of a magnet. The pole pieces were shaped in the form of a knife edge so that a very large magnetic field gradient $\frac{dy}{dx}$ was obtained. The whole apparatus was evacuated to a pressure sufficiently low to enable a Ag atom to traverse the whole length without a collision so that the mean free path was greater than the length of the apparatus. The emergent beam from the pole pieces was received on a glass plate P kept at the end of the field perpendicular to the knife edge. The beam is rendered visible on the plate by photographic techniques.



Outside the mag. field the magnetic moments of the Silver atoms were oriented in random fashion. But on entering the mag. field the magnetic moments became oriented parallel and anti-parallel to the direction of the field only and not along all possible directions as outside the field because of the space quantization. Since the atoms experienced a resultant force due to the large magnetic field gradient the direction of the force being ~~taken~~ determined by the direction which the magnetic moment had taken up w.r to the magnetic field. Thus the atoms in the two orientations move apart under the action of opposite forces & were recorded as separate deposits upon the plate.





Classically the magnetic moment M_s can have any value between $-M_s$ to $+M_s$ so that the ^{trace} On the plate should have been continuous as shown in fig (b).

The fact that two discrete equally spaced traces were found showed that the magnetic moment had two separate and discrete orientations, one with the component parallel to $\vec{B}$ and other anti parallel to $\vec{B}$. These two facts proved conclusively the existence of space quantisation for M_s .

CALCULATION :

Let : $\vec{M}_s$ = The magnetic moment of a silver atom.

$\vec{B}$ = The induction vector of the applied magnetic field.

d = The displacement of the atoms from the x-axis

The force F_x along x-axis is given by

$$F_x = - \frac{\partial}{\partial x} (-\vec{M}_s \cdot \vec{B}) = + \frac{\partial}{\partial x} (M_s B \cos \theta)$$

$$= M_s \cos \theta \frac{\partial B}{\partial x} \quad \dots \dots (1)$$

Let v be the speed parallel to the knife edge of an atom entering the mag. field

$$\therefore t = \frac{L}{v} \quad \dots \dots (2)$$

where 't' is the ^v time taken by atoms to reach the plate and 'L' is the length of the mag.

We know that:

Force = mass $\times$ acceleration

$$F_x = m_a \times a_{\text{cel}}^{\text{re}}$$

Where m_a is the mass of the α atom.

$$a_{eff}^n = \frac{F_x}{m_a}$$

putting eqⁿ (1):

$$a_{\text{eff}}^n = \frac{\mu_s \cos \theta}{m_a} \frac{\partial \theta}{\partial x} \quad (3)$$

$$\therefore S = d = \frac{1}{2} a t^2 \quad [\because S = ut + \frac{1}{2} a t^2]$$

$$= \frac{1}{2} \frac{M_s \cos \theta}{m_a} \frac{\partial \vec{B}}{\partial x} \cdot \vec{r}$$

$$= \frac{1}{2} \frac{M \cos \theta}{m_a} \frac{\partial B}{\partial x} \cdot \frac{L^2}{v^2}$$

But- $\frac{1}{v^2} = \frac{m_a}{3KT}$ (from kinetic theory)

$$\therefore d = \frac{L^2}{6KT} \mu_s \cos \theta \frac{\partial \overline{B}}{\partial x}$$

Where k is Boltzmann Constant

T is the temp. of the furnace in degree Kelvin.